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Under review as a conference paper at ICAIS 2025

COLLAB LLM: TRANSFORMING LARGE LANGUAGE MODELS INTO ACTIVE COLLABORATORS IN MULTI-TURN INTERACTIONS

Anonymous authors
Paper under double-blind review

ABSTRACT

Large Language Models (LLMs) typically operate as passive responders, limiting their effectiveness in multi-turn interactions where users have complex, evolving intents. This research introduces COLLAB LLM, a novel training framework that leverages a collaborative simulation to estimate the long-term impact of responses through Multiturn-aware Rewards (MR). By applying reinforcement learning with these rewards, COLLAB LLM encourages active intent discovery and insightful suggestions from the model, thereby transforming the nature of user-LLM interactions. We propose a multiturn interaction benchmark that includes three challenging tasks, such as collaborative document creation. Preliminary results indicate that COLLAB LLM outperforms traditional models with an average of 18.5% higher task performance and 46.3% improved interactivity, as rated by LLM judges. Furthermore, a large user study with 201 participants revealed an increase in user satisfaction by 17.6% and a reduction in time spent by 10.4%. This research aims to pave the way for more engaging and efficient AI-driven

Citation Tools

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🎓 Meta Review & Human Decision

⏳ **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

📋 AI Review from DeepReviewer will be automatically processed

📋 Summary

This paper introduces COLLAB LLM, a novel training framework designed to transform Large Language Models (LLMs) from passive responders into active collaborators in multi-turn interactions. The core idea revolves around a collaborative simulation environment and the use of Multiturn-aware Rewards (MR) to optimize LLM responses based on long-term user intent. The authors argue that traditional LLMs, trained with single-turn rewards, struggle to handle complex, evolving user intents in sustained interactions. To address this, COLLAB LLM employs a discounted cumulative gain (DCG) approach to evaluate the long-term impact of responses, encouraging the model to engage in active intent discovery and provide more insightful suggestions. The framework is evaluated through a series of experiments, including a user study with 201 participants, across tasks such as collaborative document creation, problem-solving, and interactive storytelling. The results indicate that COLLAB LLM significantly outperforms traditional models, achieving an 18.5% improvement in task performance and a 46.3% increase in interactivity, as assessed by LLM judges. Furthermore, the user study revealed a 17.6% increase in user satisfaction and a 10.4% reduction in time spent on tasks. These findings suggest that COLLAB LLM enhances user engagement and efficiency, paving the way for more dynamic and effective AI-driven conversations. The paper's contribution lies in its innovative approach to training LLMs for collaborative tasks, addressing a crucial gap in existing models' capabilities to handle complex, evolving user intents. By introducing a multiturn interaction benchmark, the authors provide a valuable tool for evaluating LLMs in more realistic scenarios. Overall, the paper presents a promising framework for advancing the field of human-AI collaboration, with significant implications for the development of more interactive and context-aware language models.

✅ Strengths

I find several aspects of this paper to be particularly strong. The core strength lies in the novel approach to training LLMs for collaborative tasks, which is a relatively unexplored area. The introduction of the COLLAB LLM framework, with its focus on long-term user intent and active collaboration, is a significant contribution. The use of a collaborative simulation environment and Multiturn-aware Rewards (MR) to optimize LLM responses is a technically innovative approach. The mathematical formulation of MR, using a discounted cumulative gain (DCG) approach, provides a clear and quantifiable method for evaluating the long-term impact of responses. The paper also introduces a multiturn interaction benchmark, which is a valuable contribution to the field. This benchmark provides a more realistic and challenging evaluation of LLMs' capabilities in collaborative scenarios, going beyond traditional single-turn tasks. The inclusion of a user study with a significant number of participants (201) strengthens the validity of the results. The reported improvements in task performance, user satisfaction, and efficiency are substantial and demonstrate the effectiveness of the proposed framework. The paper is well-structured and clearly articulates the problem, methodology, and results. The writing is clear and concise, making it accessible to a broad audience. The figures and tables effectively present the experimental results, making it easy to understand the performance of COLLAB LLM compared to the baselines. The paper's focus on transforming LLMs from passive responders to active collaborators addresses a crucial gap in existing models' capabilities, and the results suggest that this approach has the potential to significantly enhance user engagement and satisfaction in AI-driven conversations.

❌ Weaknesses

Despite the strengths of this paper, I have identified several weaknesses that warrant careful consideration. Firstly, the pap comparison with existing methods or models that address multi-turn interactions. While the 'RELATED WORK' section mentions some relevant research, it does not provide a comprehensive comparison of COLLAB LLM with specific state-of-the-art models. The baselines used in the experiments are described generally as "traditional LLMs trained with single-turn rewards," without specifying the exact models or their architectures. This lack of detailed comparison makes it difficult to fully contextualize the novelty and effectiveness of COLLAB LLM. For example, it is unclear how COLLAB LLM compares to other models that also use reinforcement learning for dialogue policy optimization or those that employ different reward mechanisms for multi-turn interactions. This omission limits my ability to fully assess the significance of the proposed framework. My confidence in this weakness is high, as the lack of specific comparisons is evident throughout the 'RELATED WORK' and 'EXPERIMENTS' sections. Secondly, while the paper provides a description of the collaborative simulation environment and a mathematical formulation for MR, it lacks sufficient details about the implementation of these components. The 'METHOD' section describes the environment as simulating real-world collaborative tasks and using a state-tracking mechanism, but it does not specify the exact algorithms or data structures used for state tracking and response adjustment. Similarly, while the formula for MR is provided, the specific implementation details of how the utility of each response is calculated are not fully explained. This lack of detail hinders the reproducibility of the method and makes it difficult to fully understand the impact of these components on the model's performance. My confidence in this weakness is also high, as the lack of implementation specifics is evident in the 'METHOD' section. Thirdly, the evaluation of the proposed framework relies heavily on LLM judges for assessing interactivity and a limited user study. While the paper mentions task completion rate as a metric, the emphasis on "interactivity, as rated by LLM judges" suggests a reliance on subjective evaluation. The user study, while involving 201 participants, could benefit from more details on the diversity of the participants. The lack of standard objective metrics for evaluating dialogue quality or task completion in multi-turn interactions, such as BLEU, ROUGE (adapted for dialogue), or metrics based on task-specific outcomes, is a significant limitation. This reliance on subjective evaluation and a limited user study weakens the validity of the results. My confidence in this weakness is high, as the reliance on LLM judges and the lack of objective metrics are clearly stated in the 'EXPERIMENTS' section. Fourthly, the paper does not discuss the potential limitations or challenges of the COLLAB LLM framework. There is no dedicated section or paragraph that addresses potential drawbacks, such as the computational cost of training, the sensitivity of the model to hyperparameter settings, or the potential for the model to generate inappropriate or biased responses. This omission provides an incomplete picture of the proposed method and its practical implications. My confidence in this weakness is high, as the absence of a limitations section is evident throughout the paper. Finally, while the paper mentions the hardware used for the experiments, it lacks details on the computational resources required for training and deploying the COLLAB LLM framework. Information on training time, the number of parameters of the model, and specific deployment challenges is missing. This lack of information makes it difficult to assess the scalability and practical feasibility of the method. My confidence in this weakness is high, as the lack of detailed computational resource information is evident in the 'EXPERIMENTS' section.

💡 Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the paper would significantly benefit from a more thorough comparison with existing approaches for multi-turn dialogue. Specifically, the authors should benchmark COLLAB LLM against state-of-the-art models that also address long-term context and user intent. This comparison should not only focus on overall performance metrics but also delve into the specific strengths and weaknesses of each approach in different scenarios. For example, how does COLLAB LLM handle situations where user intent shifts abruptly compared to models that rely on more static reward mechanisms? Including a detailed analysis of the baselines used, such as their architecture, training data, and specific configurations, is crucial for reproducibility and a deeper understanding of the results. Furthermore, the paper should clarify how the proposed framework compares to methods that use reinforcement learning for dialogue policy optimization, highlighting the unique contributions of the multiturn-aware rewards. Secondly, to enhance the reproducibility and understanding of the proposed method, the authors should provide a more detailed explanation of the collaborative simulation environment and the Multiturn-aware Rewards (MR). For the simulation environment, the paper should specify the types of interactions that are simulated, the range of user behaviors that are modeled, and the mechanisms used to generate multi-turn dialogues. For the MR, the paper should provide a clear mathematical definition of how the long-term impact of responses is estimated, including the specific metrics used to measure task completion and interactivity. It would also be beneficial to include an ablation study that examines the impact of different components of the MR on the overall performance of the model. This would help to understand the relative importance of each component and provide insights into how the MR can be further improved. The authors should also discuss the limitations of the current MR design and potential areas for future research. Thirdly, the evaluation of the proposed framework needs to be strengthened by incorporating more objective metrics and a larger, more diverse user study. While LLM judges can provide valuable insights into interactivity, they should be complemented with metrics that directly measure the quality of the generated responses, such as BLEU, ROUGE, or METEOR scores, adapted for multi-turn dialogue. The user study should include a more diverse group of participants, with varying backgrounds and levels of expertise, to ensure that the results are generalizable. The authors should also provide more details on the methodology used to conduct the user study, including the specific tasks that participants were asked to complete, the metrics used to measure user satisfaction, and the

to analyze the results. Additionally, the paper should discuss the potential limitations of the current evaluation setup and su
these limitations in future work. Fourthly, the paper should include a dedicated section discussing the potential limitations a
COLLAB LLM framework. This section should address issues such as the computational cost of training, the sensitivity of t
settings, the potential for the model to generate inappropriate or biased responses, and the challenges of deploying the model in real-world scenarios. Finally, the paper should provide more details on the computational resources required for training and deploying the COLLAB LLM framework. This should include information on training time, the number of parameters of the model, and the specific hardware used. This information is crucial for assessing the scalability and practical feasibility of the method.

❓ Questions

Based on my analysis, I have several questions that I believe are crucial for a deeper understanding of the paper. Firstly, could the authors provide more details on the computational resources required for training and deploying the COLLAB LLM framework? Specifically, what is the training time for the model, and how does the framework scale with larger models and datasets? This information is essential for assessing the practical feasibility of the method. Secondly, how does COLLAB LLM handle situations where user intent is ambiguous or shifts during the interaction? Are there any mechanisms in place to adapt to such changes in real-time? This is important for understanding the robustness of the model in real-world scenarios. Thirdly, what are the potential challenges in deploying COLLAB LLM in real-world applications, particularly in terms of maintaining context and user engagement over extended interactions? This question aims to explore the practical limitations of the framework. Fourthly, how does the proposed multiturn interaction benchmark compare to existing benchmarks in terms of complexity and realism? What specific aspects of multiturn interactions does this benchmark evaluate that others do not? This question seeks to understand the unique contributions of the proposed benchmark. Finally, could the authors elaborate on the choice of LLM judges for evaluating interactivity? How were these judges calibrated, and what measures were taken to ensure consistency in their evaluations? This question aims to understand the reliability of the subjective evaluation method.

📊 Scores

Soundness: 2.5

Presentation: 2.0

Contribution: 2.25

Rating: 3.0

AI Review from ZGCA

📋 AI Review from ZGCA will be automatically processed

📋 Summary

The paper proposes COLLAB LLM, a training framework intended to transform LLMs from passive responders into activ interactions. The key idea is a collaborative simulation that estimates the long-term impact of model responses through Mu which are then used to optimize the policy via reinforcement learning. The authors also claim a new multi-turn interaction (including collaborative document creation). Reported results include average improvements of 18.5% in task performance and 46.3% in interactivity (LLM judges), and a user study with 201 participants showing 17.6% higher satisfaction and 10.4% less time spent.

Citation Tools

✅ Strengths

- Addresses an important and under-explored challenge: proactive, multi-turn collaboration with evolving user intents (Sections 1–3).
- Conceptually appealing use of a collaborative simulation to provide long-horizon, multi-turn credit assignment via Multiturn-aware Rewards (MR) (Abstract, Section 3).
- Aspires to provide a new benchmark with tasks like collaborative document creation that could be useful for the community (Abstract, Section 4).
- Reports promising headline results: +18.5% task performance, +46.3% interactivity from LLM judges, and a 201-user study with +17.6% satisfaction and −10.4% time (Abstract, Section 4).

❌ Weaknesses

- Missing technical detail on the core contribution. Section 3 describes MR and the collaborative simulation only at a high level. There is no formal definition of MR, no reward equation, no description of how long-term impact is estimated (e.g., return formulation, discounting, horizon), no pseudo-code, and no specifics of the simulation (e.g., user model design, stochasticity, domain coverage).
- Unspecified training procedure. The RL algorithm (e.g., PPO, DPO/RLHF variants), policy/critic setup, datasets, training hyperparameters, and stability considerations are not reported (Section 3).
- Evaluation lacks essential details. The three benchmark tasks are not described (data sources, goal definitions, metrics, difficulty), and the baselines ("traditional models") are not named nor tuned or matched fairly (Section 4).
- Heavy reliance on LLM judges without validation. No details of judge models, prompts, rubric, inter-judge agreement, or correlation with human ratings are provided, risking evaluator bias and Goodharting (Abstract, Section 4).
- No statistical rigor. The paper reports mean improvements but does not provide confidence intervals, variance, or significance tests for either the automated evaluation or the 201-participant user study (Section 4).
- No ablation or failure analysis. There is no analysis isolating the contribution of MR versus the simulation or RL, nor qualitative/error analyses explaining when the approach helps or fails (Section 4).
- Related work section does not clearly differentiate MR from existing intrinsic/curiosity rewards or multi-turn RL credit assignment approaches; citations are sparse given the breadth of related literature (Section 2).
- Reproducibility concerns: No dataset/task release details, no code or model card, and insufficient methodological specification to replicate results.

❓ Questions

- Please formally define Multiturn-aware Rewards (MR): What is the reward signal at each turn? How is long-term impact estimated (e.g., discounted return, trajectory-level evaluation)? What signals are used (task success proxies, user simulator responses, auxiliary critics)?
- Describe the collaborative simulation in detail: Is the user modeled by an LLM, rules, or real logs? How is intent evolution represented? How is stochasticity handled? Provide pseudo-code for the simulation loop and data flow during training.
- What RL algorithm is used (e.g., PPO, RLHF variants, Q-learning)? What are the policy/critic architectures, training obje hyperparameters? How many training steps/samples were used?
- Precisely specify the benchmark: What are the three tasks (including collaborative document creation)? What are their evaluation metrics? Provide examples and the exact success criteria.
- List the baselines by name and configuration. How were they selected and tuned? Do they include strong multi-turn dialogue/collaboration models, RLHF-based baselines, and intrinsic/curiosity reward approaches?
- Detail the LLM-judge protocol: which model(s), prompts, rubrics, numbers of judges per sample, inter-judge agreement, calibration, and correlation with human judgments. Did you assess bias and robustness (e.g., adversarial phrasing)?
- Provide statistical analysis for all reported numbers: confidence intervals, standard errors, and significance tests per task. Also report per-task results rather than only aggregates.
- Describe the 201-participant user study: recruitment, demographics, study design (within/between-subjects), randomization, compensation, instructions, task assignment, and measurement instruments for satisfaction and time. Was IRB/ethics approval obtained?
- Include ablations: remove MR, vary the simulation fidelity, compare trajectory-level vs step-wise rewards, and test sensitivity to discount/horizon choices to illuminate what drives gains.
- Report failure cases and qualitative analyses to clarify when active collaboration helps or harms, including examples where proactivity misreads user intent.
- What safeguards are in place to prevent over-proactivity (e.g., excessive questioning, persuasive nudging), misinformation, and fairness issues across different user demographics?
- What is the computational cost of training with simulation and MR (GPU hours, carbon footprint)?

⚠️ Limitations

- Potential misalignment due to over-proactivity: optimizing for interactivity (Section 4) may increase unnecessary questions or nudging behaviors that frustrate users or bias decisions.
- Evaluator bias: heavy reliance on LLM judges without validation risks Goodharting and biased scoring; human-judge calibration is needed.
- Simulation-to-reality gap: training and evaluation in a collaborative simulation may not transfer to real users or domains if the simulator's user model is not representative (Section 3 alludes to tracking user intent but provides no details).
- Lack of transparency and reproducibility: absence of methodological specifics and resource reporting prevents scrutiny and adoption.
- Safety and societal risks: active suggestions in sensitive domains (health, finance) could amplify harms if the model confidently proposes incorrect or manipulative actions; fairness impacts across user groups are unaddressed.
- Scope and generalization: the benchmark is not described, limiting conclusions about domain breadth and robustness.

🖼️ Image Evaluation

Cross-Modal Consistency: 18/50

Textual Logical Soundness: 12/30

Visual Aesthetics & Clarity: 9/20

Overall Score: 39/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

• Visual ground truth (Figure 1)

– Sub-figure (a): "Baseline Training Loss Over Epochs" — line plot; x=Epochs (0–14), y=Loss (~1.0–1.28), single blue line v

– Sub-figure (b): "Baseline UES Over Epochs" — line plot; x=Epochs (0–14), y=UES (0–1), single orange line, highly fluctu

– Figure-level synopsis: Both panes show baseline-only dynamics; no COLLAB LLM curves or comparisons.

Citation Tools

• Major 1: Central claim ("18.5% performance, 46.3% interactivity") is said to be shown in Fig. 1, but the figure displays only baseline metrics with no comparative COLLAB LLM data. Evidence: "achieving an average of 18.5% higher task performance and 46.3% improved interactivity, as shown in Figure 1."

• Major 2: Caption promises comparative evaluation, yet no COLLAB LLM series, legends, or bars appear. Evidence: Figure 1 caption: "Evaluation of COLLAB LLM against baseline models."

• Major 3: Figure layout includes extra unlabeled panes/placeholders between (a) and (b), creating ambiguity about intended sub-figures. Evidence: Fig. 1, two unlabeled image panes between labeled (a) and (b).

• Minor 1: "UES" acronym appears in Fig. 1(b) but is not defined in text. Evidence: Fig. 1(b) label "UES"; Sec. 4 lacks definition.

• Minor 2: No dataset/task identifiers on axes/caption to map plots to experiments. Evidence: Fig. 1 has generic titles only.

2. Text Logic

• Major 1: Multiturn-aware Rewards (MR) are not formalized; no objective, components, or estimation procedure. Evidence: Sec. 3 describes MR conceptually but gives no formula/algorithm.

• Major 2: RL setup unspecified (policy initialization, optimizer, rollout policy, baselines, credit assignment, safety filters). Evidence: "By integrating MR into the model's training process…" with no training details in Sec. 3.

• Major 3: User study lacks protocol (tasks, instructions, randomization, statistical tests, rating rubric). Evidence: "a user study involving 201 participants… increase in user satisfaction by 17.6%."

• Minor 1: Abstract promises "three challenging tasks" but paper does not enumerate them. Evidence: Abstract: "includes three challenging tasks".

• Minor 2: Claims of novelty over multi-turn benchmarks not concretely contrasted. Evidence: Sec. 2 gives high-level positioning only.

3. Figure Quality

• Major 1: Core message cannot be inferred; comparison absent; duplicated/empty panes confuse reading. Evidence: Caption vs. contents of Fig. 1 do not match.

• Minor 1: Axes lack units/context (dataset, split, metric definition). Evidence: Fig. 1 axis labels are generic ("Loss", "UES").

• Minor 2: "Figure-alone" test fails; viewers cannot tell what "UES" measures or how it links to interactivity. Evidence: Fig. 1(b) lacks legend/context beyond name.

Key strengths:

- Timely problem: multi-turn collaboration and intent discovery.
- Clear high-level idea of MR and collaborative simulation.
- Promising headline improvements and sizable user study sample.

Key weaknesses:

- Central evidence missing/misaligned with Figure 1.
- Method underspecified (MR definition and RL pipeline).
- Evaluation design and statistics not reported.
- Undefined metrics (UES), unlabeled/missing sub-figures, and ambiguous caption.

📊 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 2

Soundness: 1

Presentation: 2

Contribution: 2

Rating: 3

AI Review from SafeReviewer

📋 AI Review from SafeReviewer will be automatically processed

📋 Summary

This paper introduces COLLAB LLM, a novel training framework designed to transform Large Language Models (LLMs) from passive responders into active collaborators in multi-turn interactions. The core idea revolves around a collaborative simulation environment where the LLM engages with users over multiple turns, dynamically adjusting its responses based on evolving user intent. The framework incorporates Multiturn-aware Rewards (MR), which are designed to estimate the long-term impact of the LLM's responses, moving beyond single-turn evaluation metrics. By integrating MR into a reinforcement learning framework, the authors aim to optimize the LLM's ability to proactively discover user intents and provide insightful suggestions. The paper presents preliminary experimental results on a newly proposed benchmark featuring tasks such as collaborative document creation. These results suggest that COLLAB LLM outperforms traditional models in terms of task performance and interactivity, as evaluated by LLM judges. Furthermore, a user study with 201 participants indicates increased user satisfaction and reduced time spent on tasks when using COLLAB LLM. The authors position their work as a significant step towards more engaging and efficient AI-driven conversations, with the potential to enhance user experience in complex, multi-turn interactions. The paper's significance lies in its attempt to address the limitations of current LLMs in collaborative settings by introducing a framework that explicitly optimizes for long-term interaction quality. However, the paper's brevity and lack of detail, likely due to the constraints of a concise submission format, make it difficult to fully assess the validity and generalizability of the presented results. The absence of specific details regarding the implementation of the collaborative simulation environment, the calculation of the Multiturn-aware Rewards, and the architecture of the baseline models used for comparison, hinders a thorough evaluation of the proposed methodology. D
paper's core idea of using reinforcement learning with multi-turn rewards to enhance collaborative interactions is pron
investigation.

✅ Strengths

The core strength of this paper lies in its innovative approach to transforming LLMs into active collaborators through the use of a reinforcement learning framework guided by Multiturn-aware Rewards (MR). The concept of MR, designed to estimate the long-term impact of responses, is a significant contribution, as it moves beyond the typical single-turn evaluation metrics commonly used in LLM research. This focus on long-term interaction quality is particularly relevant for collaborative tasks where the effectiveness of an LLM depends on its ability to understand and adapt to evolving user intents over multiple turns. The introduction of a collaborative simulation environment, where the LLM interacts with users over multiple turns, is another notable strength. This environment allows for the dynamic tracking of user intent and enables the model to adjust its responses accordingly, fostering a more interactive and engaging user experience. The preliminary results, which suggest that COLLAB LLM outperforms traditional models in terms of task performance and interactivity, as evaluated by LLM judges, are also encouraging. Furthermore, the user study, involving 201 participants, provides initial evidence of increased user satisfaction and reduced time spent on tasks when using COLLAB LLM. These findings, while preliminary, indicate the potential of the proposed framework to enhance user experience in complex, multi-turn interactions. The paper's focus on collaborative tasks, such as document creation, addresses a practical and important application area for LLMs, further highlighting the significance of this work. The authors' attempt to address the limitations of current LLMs in collaborative settings by explicitly optimizing for long-term interaction quality is a valuable contribution to the field. The paper's core idea of using reinforcement learning with multi-turn rewards to enhance collaborative interactions is promising and warrants further investigation.

❌ Weaknesses

The most significant weakness of this paper is the lack of detailed information regarding the implementation of the proposed framework, which makes it difficult to fully understand and evaluate the presented results. Specifically, the paper does not provide sufficient details about the collaborative simulation environment, the calculation of the Multiturn-aware Rewards (MR), the architecture of the baseline models used for comparison, or the specific reinforcement learning algorithm employed. This lack of detail is a recurring theme across multiple valid critical perspectives. For instance, the paper mentions a "collaborative simulation environment" (Line 72) but fails to describe its specific components, such as the user model or the simulation process. This omission makes it unclear how the multi-turn interactions are simulated and how the user intent is tracked across conversations. The absence of this information hinders the reproducibility of the experiments and limits the ability of other researchers to build upon this work. The paper introduces "Multiturn-aware Rewards (MR)" (Line 73) but does not provide the specific mathematical formulation or the features used to estimate the long-term impact of responses. This lack of clarity makes it difficult to understand how the rewards are calculated and whether they effectively capture the desired long-term interaction quality. The paper states that MR is "designed to estimate the long-term impact of the model's responses" (Line 73), but without the specific details, it is impossible to assess the validity of this claim. The paper mentions using "reinforcement learning" (Line 74) but does not specify the exact algorithm used, such as policy gradients or Q-learning. This lack of information makes it difficult to understand how the model is trained and optimized. Furthermore, the paper does not provide details about the architecture of the baseline models used for comparison. The paper compares COLLAB LLM to "traditional LLMs and baseline models" (Line 88) but does not specify which models were used or whether they were fine-tuned. This lack of information makes it difficult to assess the fairness of the comparison and the significance of the reported improvements. The paper reports quantitative results, such as "18.5% higher task performance" (Line 85) and "46.3% improved interactivity" (Line 85), but it is unclear how these metrics were calculated. The paper mentions that interactivity was "rated by LLM judges" (Line 85), but it does not specify the criteria used for these ratings or the specific LLM used as a judge. This lack of transparency makes it difficult to assess the validity of these metrics and the reported improvements. The paper also lacks details about the user study, such as the demographics of the participants, the specific tasks they performed, and the metrics used to measure user satisfaction. While the paper mentions a user study with "201 participants" (Line 86), it does not provide sufficient details to assess the validity and generalizability of the findings. The paper's brevity, likely due to the constraints of a concise submission format, exacerbates these issues. The lack of an appendix further limits the amount of detail that can be provided. These limitations, while understandable, significantly impact the paper's credibility and the ability of other researchers to replicate and build upon this work. The absence of specific details regarding the implementation of the collaborative simulation environment, the calculation of the Multiturn-aware Rewards, and the architecture of the baseline models used for comparison, hinders a thorough evaluation of the propose

Citation Tools

clarity surrounding the evaluation metrics and the user study further compounds these issues. Overall, the lack of detailed to fully assess the validity and generalizability of the presented results, which is a major weakness of this paper.

💡 Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. First and foremost, the authors should provide a more detailed description of the collaborative simulation environment. This should include a clear explanation of the user model, the simulation process, and how user intent is tracked across multiple turns. Specifically, the authors should describe the algorithms or techniques used to simulate user behavior and how the environment adapts to the LLM's responses. This level of detail is crucial for understanding the experimental setup and for enabling other researchers to replicate the results. Second, the authors should provide a precise mathematical formulation of the Multiturn-aware Rewards (MR). This should include a clear definition of the features used to estimate the long-term impact of responses and the specific calculations used to combine these features into a single reward value. The authors should also explain the rationale behind the choice of these features and how they relate to the desired long-term interaction quality. This level of detail is essential for understanding the core mechanism of the proposed framework. Third, the authors should explicitly state the reinforcement learning algorithm used for training the LLM. This should include details about the specific variant of the algorithm, the hyperparameters used, and the optimization process. The authors should also explain why they chose this particular algorithm and how it is suited to the proposed task. This level of detail is necessary for understanding the training process and for enabling other researchers to replicate the results. Fourth, the authors should provide detailed information about the architecture and training procedures of the baseline models used for comparison. This should include the specific models used, whether they were fine-tuned, and the details of any fine-tuning process. This level of detail is crucial for assessing the fairness of the comparison and the significance of the reported improvements. Fifth, the authors should provide a clear explanation of how the evaluation metrics, such as task performance and interactivity, were calculated. This should include the specific formulas used, the criteria used by the LLM judges, and the specific LLM used as a judge. The authors should also justify the choice of these metrics and explain how they relate to the desired long-term interaction quality. Sixth, the authors should provide more details about the user study, including the demographics of the participants, the specific tasks they performed, and the metrics used to measure user satisfaction. This level of detail is necessary for assessing the validity and generalizability of the findings. Finally, given the brevity of the main paper, the authors should consider including an appendix with additional details about the implementation, the experiments, and the results. This would provide a valuable resource for other researchers and would enhance the credibility of the work. By addressing these points, the authors can significantly improve the clarity, reproducibility, and impact of their research.

❓ Questions

Based on my analysis, several key questions remain unanswered, primarily concerning the methodological choices and the evaluation process. First, I am curious about the specific design choices behind the collaborative simulation environment. What specific algorithms or techniques were used to simulate user behavior, and how was the environment designed to adapt to the LLM's responses? Understanding the underlying mechanisms of the simulation environment is crucial for assessing the validity of the experimental results. Second, I would like to know more about the rationale behind the specific features chosen for the Multiturn-aware Rewards (MR). What criteria were used to select these features, and how do they relate to the desired long-term interaction quality? A deeper understanding of the reward function is essential for evaluating the effectiveness of the proposed framework. Third, I am interested in the specific reasons for choosing the particular reinforcement learning algorithm used for training the LLM. What were the alternatives considered, and why was the chosen algorithm deemed most suitable for this task? Understanding the rationale behind this choice is important for assessing the validity of the training process. Fourth, I would like to know more about the architecture and training procedures of the baseline models used for comparison. Were these models fine-tuned, and if so, what were the details of the fine-tuning process? A clear understanding of the baseline models is crucial for assessing the significance of the reported improvements. Fifth, I am curious about the specific criteria used by the LLM judges to rate interactivity. What aspects of the interaction were they evaluating, and how were these aspects quantified? A deeper understanding of the evaluation process is necessary for assessing the validity of the reported improvements in interactivity. Finally, I would like to know more about the demographics of the participants in the user study and the specific tasks they performed. This information is crucial for assessing the generalizability of the findings. Addressing these questions would provide a more complete understanding of the proposed framework and its evaluation, and would help to clarify the contributions of this work.

📊 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Confidence: 4.0

Rating: 1.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights

Citation Tools